Highly-Expressive Spaces of Well-Behaved Transformations: Keeping It Simple

Supplemental Material

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Contents

In addition to the **attached video** (also available online at https://www.youtube.com/watch?v=d5yNXE5mwcQ), this supplemental material contains:

1. Additional results: a rectangle-to-horseshoe transformation; a 3D example involving meshes; computation time for warping a 3D volume; an example for handling a large set of landmarks; fitting a CDF using a CPAB transformation.
2. Visualization of a CPA basis in the 2D case.
3. Proofs for the lemmas and theorems.
4. Accuracy analysis for the proposed solver.
5. A discussion on certain differential-geometric aspects of the proposed representation.

1. Additional Select Results

1.1. A Rectangle-to-horseshoe Transformation

See Figure 1.

1.2. 3D

Figure 2 shows $T^{\boldsymbol{\theta}}$, sampled from the prior, applied to a 3D shape. Here, $T^{\boldsymbol{\theta}} \in M^{\partial,\mathrm{vp}}$, where $\mathcal{P}$ consists of $N_c = 320 = 4^3 \times 5$ cells (we used a layout of 4^3 cubes, and then divided each one into 5 tetrahedra) and $N_v = 125$. The dimension of the transformation is $d = 20$ (and not, *e.g.*, $375 = 3N_v$) due to the boundary-condition and volume-preserving constraints.

In terms of timing, with our current GPU implementation, transforming a volume with 100^3 voxels, where $\mathcal{P}$ consists of $N_c = 320 = 4^3 \times 5$ cells and with boundary conditions (and no volume preservation) (so $N_c = 320 = 4^3 \times 5$, $N_v = 125$, and $d = 225$), takes about 0.2 [sec] on average.

1.3. Handling a Large Number of Landmarks

In the hand experiments in the paper we show inference for landmark-based warping using a relatively small set of landmarks on real data. The inference technique used there was MAP estimation using the Metropolis algorithm. Here we show, using synthetic data, that the framework can easily utilize a large set of landmark for inference over the latent transformation between the landmark sets; see Fig. 3. The technique used here is ML estimation using the Metropolis algorithm. As an aside remark, it is unsurprising that with more data the results improve, suggesting some empirical evidence that the statistical estimation procedure is *consistent*. Our point here, however, is that most other methods cannot utilize a large set of landmarks so easily.

[1] http://tosca.cs.technion.ac.il

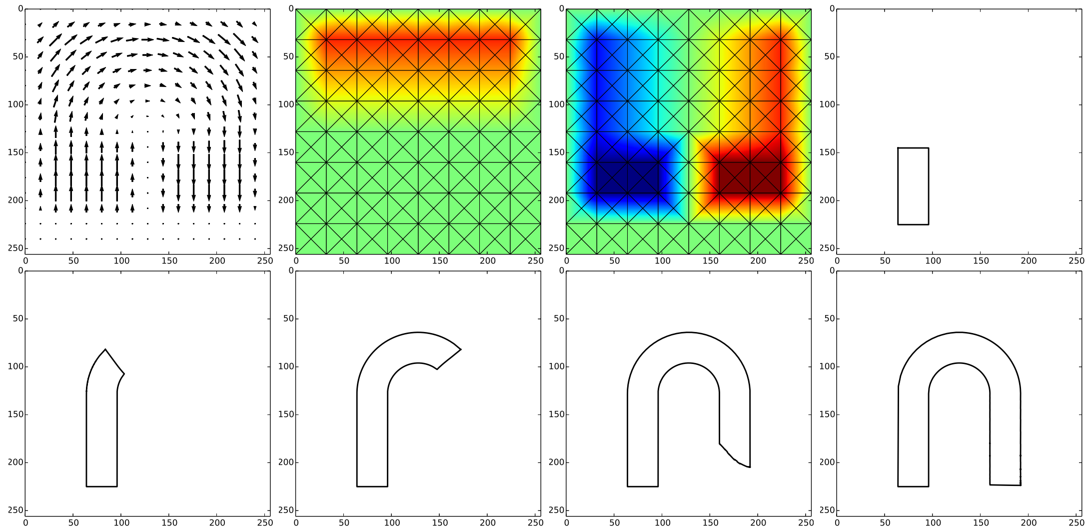


Figure 1: Top left to bottom right: $\boldsymbol{v}^{\boldsymbol{\theta}}$, its horizontal and vertical components, and 5 steps of a rectangle-to-horseshoe process. The lower bar of the rectangle does not move since $\boldsymbol{v}^{\boldsymbol{\theta}} = \mathbf{0}$ in this location, while the upper bar eventually stops when the horseshoe is completed (for the same reason).

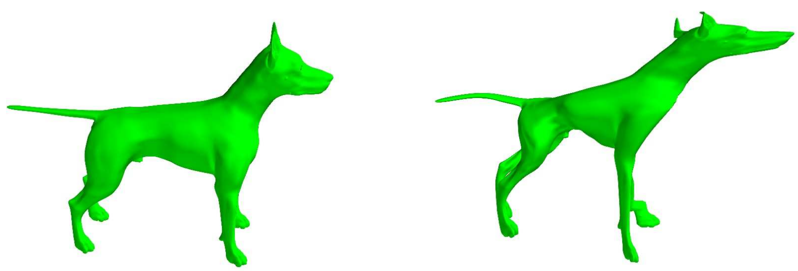


Figure 2: A 3D example. Left: Source shape (taken from the Tosca Shape Dataset[1]). Right: Result of applying a volume-preserving transformation, sampled from the prior, to the source.

1.4. Representing a CDF using a CPAB Transformation

In Fig. 4 we show inference results (using the Metropolis algorithm) for fitting $F^{\boldsymbol{\theta}} \triangleq T^{\boldsymbol{\theta}} \circ F_0$ (where F_0 is the CDF of the uniform distribution on $J = [-7, 7]$) to some nominal CDF – in this case the CDF of a 3-component Gaussian mixture. As the figure shows, $F^{\boldsymbol{\theta}}$ well approximates the target CDF.

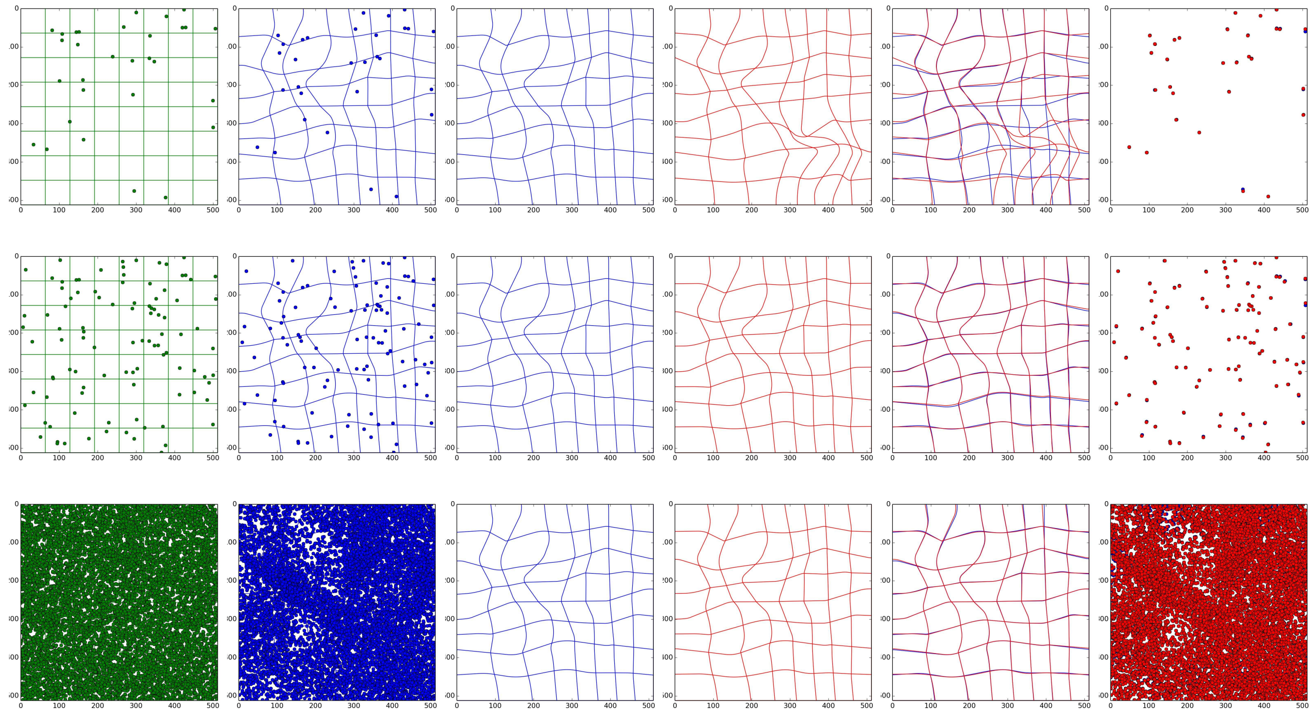
2. Visualization of a CPA Basis in the 2D Case

Figure 5 depicts the CPA fields that form a basis for $\mathcal{V}$ (a 2D case and a nominal tessellation).

3. Proofs

In what follows, the operations of multiplication of a map by a scalar and addition of two maps are defined in the standard way; *i.e.*, if f and g are two $\Omega \to \mathbb{R}^n$ maps and $\alpha \in \mathbb{R}$, we define $(f+g) : \boldsymbol{x} \mapsto f(\boldsymbol{x}) + g(\boldsymbol{x})$ and $(\alpha f) : \boldsymbol{x} \mapsto \alpha f(\boldsymbol{x})$.

Proof of Lemma 1. **First**, we prove $\mathcal{V}'$ is linear by showing its closure under linear combinations. Let $\alpha \in \mathbb{R}$ and let f and f' be two PA maps, where $f : \boldsymbol{x} \mapsto A_{\gamma(\boldsymbol{x})}\widetilde{\boldsymbol{x}}$ and $f' : \boldsymbol{x} \mapsto A'_{\gamma(\boldsymbol{x})}\widetilde{\boldsymbol{x}}$. Now note that $\alpha f : \boldsymbol{x} \mapsto \alpha \left(A_{\gamma(\boldsymbol{x})}\widetilde{\boldsymbol{x}}\right) = \left(\alpha A_{\gamma(\boldsymbol{x})}\right)\widetilde{\boldsymbol{x}}$ and $f + f' : \boldsymbol{x} \mapsto A_{\gamma(\boldsymbol{x})}\widetilde{\boldsymbol{x}} + A'_{\gamma(\boldsymbol{x})}\widetilde{\boldsymbol{x}} = \left(A_{\gamma(\boldsymbol{x})} + A'_{\gamma(\boldsymbol{x})}\right)\widetilde{\boldsymbol{x}}$ are PA maps, as follows from the linearity of $\mathbb{R}^{n\times(n+1)}$. **Second**, since $\mathcal{V}$ is the intersection of two linear spaces, $\mathcal{V}$ and the space of $\Omega \to \mathbb{R}^n$ continuous maps, it follows that $\mathcal{V}$ is a linear



(a) src: grid lines & landmarks (b) dst: grid lines & landmarks (c) dst: grid lines (d) $T^{\boldsymbol{\theta}}$(dst): grid lines (e) dst and $T^{\boldsymbol{\theta}}$(dst): grid lines (f) dst and $T^{\boldsymbol{\theta}}$(dst): landmarks

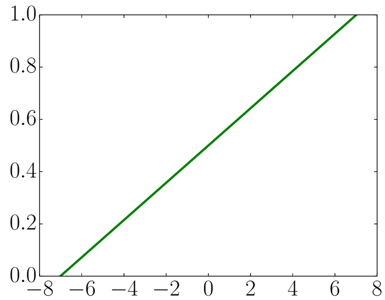
Figure 3: Maximum Likelihood (ML) estimates for latent transformations between sets of landmarks. The dimension of the transformation space that was used in this example was $d = 38$. Rows differ only in the number of landmarks, N_{pts}, being used. Top row: $N_{\text{pts}} = 30$. Middle row: $N_{\text{pts}} = 100$. $N_{\text{pts}} = 10,000$. Colors: green=src, blue=dst, red=$T^{\boldsymbol{\theta}}$(src), where $T^{\boldsymbol{\theta}}$ is the ML transformation, computed using the Metropolis algorithm. (a) original grid lines + src landmarks; (b) grid lines deformed using the sought-after transformation + dst landmarks. (c) grid lines deformed using the sought-after transformation. (d) grid lines deformed using the ML transformation. (e) grid lines deformed using the sought-after transformation (blue) and grid lines deformed using the ML transformation (red). (f) dst landmarks (blue) and the src landmarks transformed using the ML transformation (red).

space too. **Third**, that $D \triangleq \dim(\mathcal{V}) = (n^2 + n) \times N_c$ is trivial since any element of $\mathcal{V}'$ is defined by N_c (unconstrained) matrices of size $n \times (n+1)$. **Finally**, since we restricted the discussion to intervals, triangles, and tetrahedra, the values a CPA map takes at the vertices in a given cell uniquely define the A of that cell (and as we will see in the proof of Lemma 2, continuity of the field across inter-cell boundaries follows from the continuity at the vertices). Since in each one of these N_v vertices there are n degrees of freedom, it follows that $d \triangleq \dim(\mathcal{V}) = n \times N_v$. $\square$

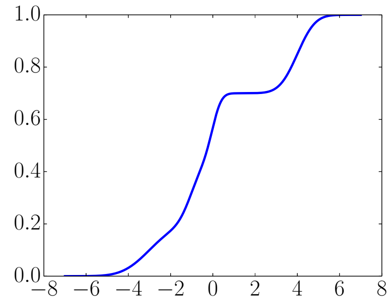
For concreteness, we show here are the details for the case $n = 2$, the other cases being essentially identical. Let $\boldsymbol{x}_1$, $\boldsymbol{x}_2$ and $\boldsymbol{x}_3$ denote the vertices of a (non-degenerate) triangle, and let $\boldsymbol{v}_1$, $\boldsymbol{v}_2$ and $\boldsymbol{v}_3$ denote the values the CPA velocity fields takes at these vertices, respectively. Letting $A = \left[\begin{smallmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{smallmatrix}\right]$ denote the matrix associated with this triangle, we have

$$\underbrace{\begin{bmatrix} \boldsymbol{v}_1 \\ \boldsymbol{v}_2 \\ \boldsymbol{v}_3 \end{bmatrix}}_{6\times 1} = \underbrace{\begin{bmatrix} A\widetilde{\boldsymbol{x}}_1 \\ A\widetilde{\boldsymbol{x}}_2 \\ A\widetilde{\boldsymbol{x}}_3 \end{bmatrix}}_{6\times 1} = \underbrace{\begin{bmatrix} \widetilde{\boldsymbol{x}}_1^T & \boldsymbol{0}_{1\times 3} \\ \boldsymbol{0}_{1\times 3} & \widetilde{\boldsymbol{x}}_1^T \\ \widetilde{\boldsymbol{x}}_2^T & \boldsymbol{0}_{1\times 3} \\ \boldsymbol{0}_{1\times 3} & \widetilde{\boldsymbol{x}}_3^T \\ \widetilde{\boldsymbol{x}}_3^T & \boldsymbol{0}_{1\times 3} \\ \boldsymbol{0}_{1\times 3} & \widetilde{\boldsymbol{x}}_3^T \end{bmatrix}}_{6\times 6} \begin{bmatrix} a_{11} \\ a_{12} \\ a_{13} \\ a_{21} \\ a_{22} \\ a_{23} \end{bmatrix} \Longrightarrow \begin{bmatrix} a_{11} \\ a_{12} \\ a_{13} \\ a_{21} \\ a_{22} \\ a_{23} \end{bmatrix} = \begin{bmatrix} \boldsymbol{v}_1 \\ \boldsymbol{v}_2 \\ \boldsymbol{v}_3 \end{bmatrix} \begin{bmatrix} \widetilde{\boldsymbol{x}}_1^T & \boldsymbol{0}_{1\times 3} \\ \boldsymbol{0}_{1\times 3} & \widetilde{\boldsymbol{x}}_1^T \\ \widetilde{\boldsymbol{x}}_2^T & \boldsymbol{0}_{1\times 3} \\ \boldsymbol{0}_{1\times 3} & \widetilde{\boldsymbol{x}}_3^T \\ \widetilde{\boldsymbol{x}}_3^T & \boldsymbol{0}_{1\times 3} \\ \boldsymbol{0}_{1\times 3} & \widetilde{\boldsymbol{x}}_3^T \end{bmatrix}^{-1} \tag{1}$$

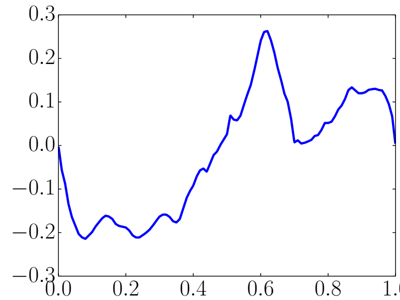
(the matrix is indeed invertible since the 3 vertices of a non-degenerate triangle are not co-linear).



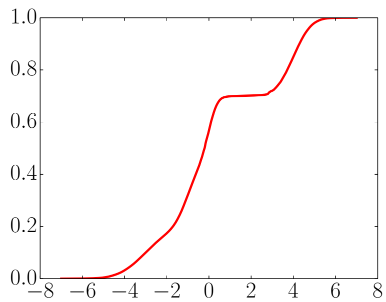
(a) F_0 – the source CDF



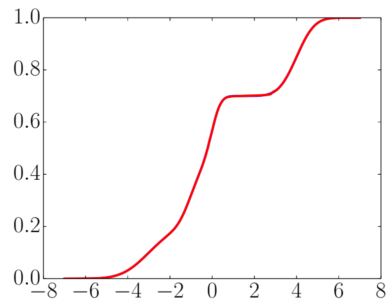
(b) F – the target CDF



(c) The estimated $\boldsymbol{v}^{\boldsymbol{\theta}}$



(d) $T^{\boldsymbol{\theta}} \circ F_0$



(e) $T^{\boldsymbol{\theta}} \circ F_0$ (red) overlaid on F (blue

Figure 4: Representing a CDF using a CPAB transformation. The source CDF is the CDF of the uniform distribution. The target is the CDF of a 3-component Gaussian mixture. The setting and the inference procedure, where here we used a Maximum Likelihood (ML) solution obtained using MCMC, are virtually the same as in the monotonic-regression experiment in the paper, except here we also used simulated annealing, gradually reducing the standard deviation of the "noise" to zero. Note the inferred velocity field is not so smooth, as is expected from a ML solution in this case. MAP estimation (not shown) leads to a smoother field, analogously to the results shown in the paper for monotonic regression.

Proof of Lemma 2. For concreteness, we prove it for $n = 2$. The other cases are handled similarly. Let $n = 2$ and let $\boldsymbol{v}_{\boldsymbol{A}} \in \mathcal{V}'$, where $\boldsymbol{A} = (A_1, \ldots, A_{N_c})$. While $\boldsymbol{v}_{\boldsymbol{A}}$ is continuous on every cell, in general it is discontinuous on inter-cell boundaries. Consider two adjacent cells, U_i and U_j. Let us denote their 2 shared vertices by $\boldsymbol{x}_a$ and $\boldsymbol{x}_b$ and let A_i and A_j denote the corresponding 2×3 matrices. Continuity of $\boldsymbol{v}_{\boldsymbol{A}}$ at $\boldsymbol{x}_a$ requires 2 (more generally, n) linear constraints on the values of A_i and A_j: one for the horizontal component of $\boldsymbol{v}_{\boldsymbol{A}}$ and one for the vertical component of $\boldsymbol{v}_{\boldsymbol{A}}$. Similarly, continuity at $\boldsymbol{x}_b$ requires another constraint pair. Thus, the continuity at both $\boldsymbol{x}_a$ and $\boldsymbol{x}_b$ implies the following 4 linear constraints:

$$\underbrace{\begin{bmatrix} \tilde{\boldsymbol{x}}_a^T & \mathbf{0}_{1\times 3} & -\tilde{\boldsymbol{x}}_a^T & \mathbf{0}_{1\times 3} \\ \mathbf{0}_{1\times 3} & \tilde{\boldsymbol{x}}_a^T & \mathbf{0}_{1\times 3} & -\tilde{\boldsymbol{x}}_a^T \\ \tilde{\boldsymbol{x}}_b^T & \mathbf{0}_{1\times 3} & -\tilde{\boldsymbol{x}}_b^T & \mathbf{0}_{1\times 3} \\ \mathbf{0}_{1\times 3} & \tilde{\boldsymbol{x}}_b^T & \mathbf{0}_{1\times 3} & -\tilde{\boldsymbol{x}}_b^T \end{bmatrix}}_{4\times 12} \underbrace{\begin{bmatrix} \mathrm{vec}(A_i) \\ \mathrm{vec}(A_j) \end{bmatrix}}_{12\times 1} = \mathbf{0}_{4\times 1} \text{ where } \mathrm{vec}\left(\left[\begin{smallmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{smallmatrix}\right]\right) \triangleq \begin{bmatrix} a_{11} \\ a_{12} \\ a_{13} \\ a_{21} \\ a_{22} \\ a_{23} \end{bmatrix} . \tag{2}$$

Continuity of $\boldsymbol{v}_{\boldsymbol{A}}$ at both $\boldsymbol{x}_a$ and $\boldsymbol{x}_b$ implies its continuity throughout their join, since for of the two affine maps, the value at a point on the join is a convex combination of the values at $\boldsymbol{x}_a$ and $\boldsymbol{x}_b$; i.e., $A_i \widetilde{\boldsymbol{x}}_a = A_j \widetilde{\boldsymbol{x}}_a$ and $A_i \widetilde{\boldsymbol{x}}_b = A_j \widetilde{\boldsymbol{x}}_b$ imply

$$A_i(\lambda \widetilde{\boldsymbol{x}}_a + (1-\lambda)\widetilde{\boldsymbol{x}}_b) = A_j(\lambda \widetilde{\boldsymbol{x}}_a + (1-\lambda)\widetilde{\boldsymbol{x}}_b) \;\forall \lambda \in [0,1] \,. \tag{3}$$

Any $\boldsymbol{v}_{\boldsymbol{A}}$ whose A_i and A_j satisfy this linear system is thus continuous on $U_i \cup U_j$. Similar constraints may be enforced for

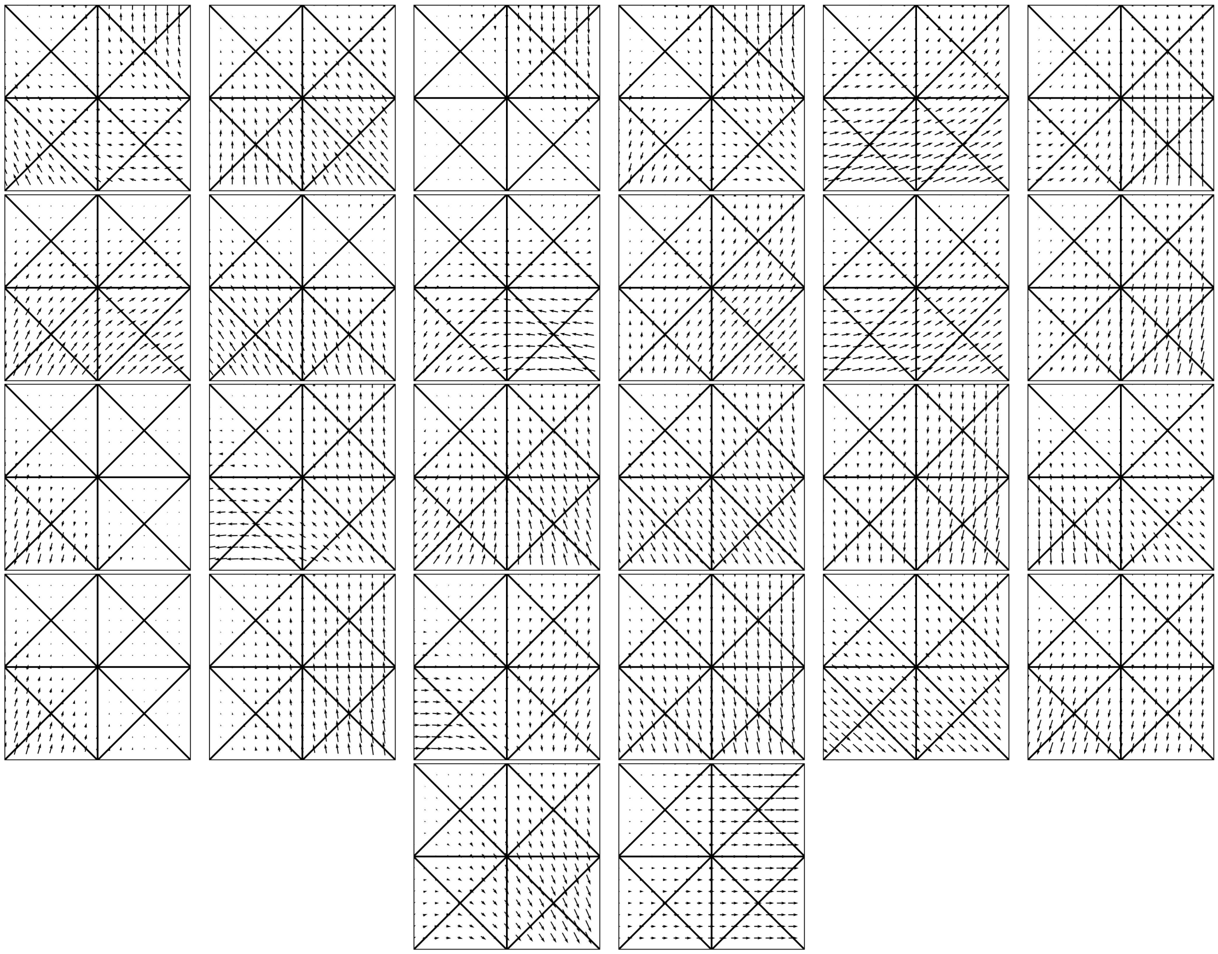


Figure 5: The vector fields that constitute a 26-dimensional orthonormal basis of $\mathcal{V}$, obtained by SVD on B

other pairs of adjacent cells, and can be stacked together in an analogous equation:

$$\underset{4N_e \times 6N_c}{L} \underset{6N_c \times 1}{\mathbf{vec}(\boldsymbol{A})} = \mathbf{0}_{4N_e \times 1} \qquad \mathbf{vec}(\boldsymbol{A}) \triangleq \begin{bmatrix} \mathrm{vec}(A_1) \\ \vdots \\ \mathrm{vec}(A_{N_c}) \end{bmatrix} \in \mathbb{R}^D = \mathbb{R}^{6N_c} \tag{4}$$

where N_e is the number of shared line segments in $\mathcal{P}$ and L is the constraint matrix. Any $\boldsymbol{v_A}$ whose $\boldsymbol{A}$ satisfies this linear system is thus everywhere continuous. We conclude that the null space of L, denoted by $\mathrm{null}(L)$, coincides with $\mathcal{V}$. □

Recall that the columns of $\boldsymbol{B} = [\, \boldsymbol{B}_1 \; \dots \; \boldsymbol{B}_d \,] \in \mathbb{R}^{D \times d}$ denote the orthonormal basis of $\mathrm{null}(L)$ obtained via SVD.

Proof of Lemma 3. The matrix associated with $L_{\boldsymbol{v}^{\boldsymbol{\theta}}_{\mathrm{vert}} \mapsto \boldsymbol{\theta}}$ is a transition matrix between two known $d-$dimensional bases. □

The reader may wonder why, given that a basis for $\mathcal{P}_{\mathrm{vert}}$ is so easy to define (*i.e.*, take the standard basis for $\mathcal{P}_{\mathrm{vert}} \cong \mathbb{R}^{N_v \times n}$), we bother to go thorough the process of creating $\boldsymbol{B}$ (*i.e.*, creating L and computing SVD of its null space). The answer is threefold. **First**, for CPA velocity fields, constraints such as zero traces (that are related to volume preservation) or the ones needed for extending CPA fields outside Ω are much easier to define via the null-space approach: we just need to add more rows to L, as we will see shortly. **Second**, we usually prefer to define our smoothness priors on the A's, rather than the values they generate at the vertices; *e.g.*, suppose a CPA field is purely affine. In which case, the $A's$ are all the same, and w.r.t. the bB basis the field is very smooth (we mean smoothness in the machine-learning sense, not calculus one, although in this particular example both apply). However, w.r.t. the vertex-based basis, the field is not so smooth since the velocities

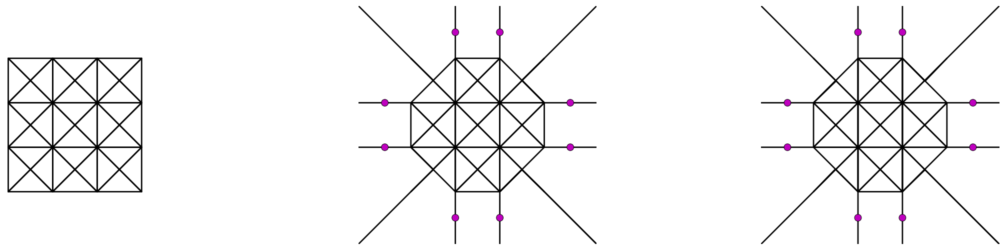


Figure 6: Left: a triangular tessellation on compact region (a square in this case). Middle: Extending it to the whole of $\mathbb{R}^2$. The circles stand for auxiliary vertices where continuity is enforced (in addition to the continuity constraints at the original vertices, including the corners of the square). Right: Using the color scheme from the paper, and the extended tessellation, we show the horizontal component of some velocity field which is CPA on the whole of $\mathbb{R}^2$ (due to the additional constraints).

change with the vertex location. **Third**, $\boldsymbol{B}$ provides an easy way to project priors from $\mathcal{V}'$ to $\mathcal{V}$ (and similarly for the other subspaces mentioned in the paper).

Proof of Lemma 4. We again focus on the case where $n=2$, the other cases being similar. To enforce zero traces, we add to L rows derived from the following constraints:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \operatorname{vec}(A_c) = 0 \qquad\qquad \forall\, c \in \{1, \ldots, N_c\}\,. \tag{5}$$

Likewise, to nullify, *e.g.*, the horizontal component of the velocity across the rightmost boundary of $\Omega = [0, x_{\text{max}}] \times [0, y_{\text{max}}]$ we add rows derived from the constraints

$$\begin{bmatrix} \widetilde{\boldsymbol{x}}_a^T & \mathbf{0}_{1\times 3} \end{bmatrix} \operatorname{vec}(A_c) = 0 \qquad\qquad \forall \tag{6}$$

for every vertex $\boldsymbol{x}_a$ of some cell U_c such that $\boldsymbol{x}$ is on the rightmost boundary of Ω. □

Proof of Lemma 5. In case $n=1$ and $\Omega = [x_{\text{xmin}}, x_{\text{xmax}}]$, we extend the leftmost and rightmost intervals to $-\infty$ and ∞, respectively, with adding no cells or constraints on $\boldsymbol{v}^{\boldsymbol{\theta}}$. If Ω is a 2D rectangle, we extend the outer cells of $\mathcal{P}$ (rendering them non-triangular) to cover the whole of $\mathbb{R}^2$ and add certain additional continuity constrains on $\boldsymbol{v}^{\theta}$. The process is best explained by the example shown in Fig. 6. By an argument similar to the one used earlier to describe how continuity at two points ensures continuity of a PA field on their join, this process ensures the extended field is CPA over the whole of $\mathbb{R}^2$. □

Mathematically, the case with $n=3$ can be handled in a similar way to the case with $n=2$. However, unlike all the other 3D-related options mentioned in the paper, we did not implement this option for the 3D case. Rather, in the 3D case we opted to ensure the CPA property by imposing boundary constraints.

Before proceeding to the proofs of the theorems, recall that the solution, $t \mapsto \psi^t_{\boldsymbol{\theta},c}(\boldsymbol{x})$, to an ODE with an $\Omega \to \mathbb{R}^n$ *affine* velocity field, $\boldsymbol{x} \mapsto A_{c,\boldsymbol{\theta}}\widetilde{\boldsymbol{x}}$, is

$$\begin{bmatrix} \psi^t_{\boldsymbol{\theta},c}(\boldsymbol{x}) \\ 1 \end{bmatrix} \triangleq T_{A_{c,\boldsymbol{\theta}},t}\widetilde{\boldsymbol{x}}\,, \qquad A_{c,\boldsymbol{\theta}} \in \mathbb{R}^{n\times(n+1)}\,, \qquad T_{A_{c,\boldsymbol{\theta}},t} \triangleq \mathbf{expm}(t\widetilde{A_{c,\boldsymbol{\theta}}})\,, \qquad \widetilde{A_{c,\boldsymbol{\theta}}} = \begin{bmatrix} A_{c,\boldsymbol{\theta}} \\ \mathbf{0}_{1\times n} \end{bmatrix}. \tag{7}$$

Also recall that $\boldsymbol{\phi}^{\boldsymbol{\theta}}(\boldsymbol{x},t)$ is given by the concatenation of such solutions

$$\boxed{\boldsymbol{\phi}^{\boldsymbol{\theta}}(\boldsymbol{x},t) = (\psi^{t_m}_{\boldsymbol{\theta},c_m} \circ \ldots \circ \psi^{t_2}_{\boldsymbol{\theta},c_2} \circ \psi^{t_1}_{\boldsymbol{\theta},c_1})(\boldsymbol{x})}\,, \tag{8}$$

that $\exp(\boldsymbol{v}^{\theta}) \triangleq T^{\boldsymbol{\theta}}$ is defined via $T^{\boldsymbol{\theta}}(\boldsymbol{x}) = \boldsymbol{\phi}^{\boldsymbol{\theta}}(\boldsymbol{x},1)$, and that the compact notation in the boxed equation above hides an important detail: the number of the trajectory segments, their durations, and the cells involved (where a cell may appear more than once), *all depend on* $\boldsymbol{x}$. They all also depend on $\boldsymbol{\theta}$ and t, except the first cell; *i.e.*, $c_1 = \gamma(\boldsymbol{x})$; *i.e.*,

$$\boldsymbol{\phi}^{\boldsymbol{\theta}}(\boldsymbol{x},t) = (\psi^{t_{m_{\boldsymbol{x},\boldsymbol{\theta},t}}(\boldsymbol{x},\boldsymbol{\theta},t)}_{\boldsymbol{\theta},c_{m_{\boldsymbol{x},\boldsymbol{\theta},t}}(\boldsymbol{x},\boldsymbol{\theta})} \circ \ldots \circ \psi^{t_2(\boldsymbol{x},\boldsymbol{\theta},t)}_{\boldsymbol{\theta},c_2(\boldsymbol{x},\boldsymbol{\theta})} \circ \psi^{t_1(\boldsymbol{x},\boldsymbol{\theta},t)}_{\boldsymbol{\theta},\gamma(\boldsymbol{x})})(\boldsymbol{x})\,. \tag{9}$$

Proof of Theorem 1. **Part (i).** For a given $\boldsymbol{x}$, the map $\boldsymbol{\theta} \mapsto \boldsymbol{v}^{\boldsymbol{\theta}}(\boldsymbol{x})$ is linear. Thus, $\boldsymbol{v}^{\boldsymbol{\theta}} = \boldsymbol{v}^{\alpha\boldsymbol{\theta}}/\alpha$. Moreover,

$$\xi^{\boldsymbol{\theta}}(\boldsymbol{x},t) \triangleq \boldsymbol{\phi}^{\alpha\boldsymbol{\theta}}(\boldsymbol{x},t/\alpha) = \boldsymbol{x} + \int_0^{t/\alpha} \boldsymbol{v}^{\alpha\boldsymbol{\theta}}(\boldsymbol{\phi}^{\alpha\boldsymbol{\theta}}(\boldsymbol{x},\tau))\,d\tau = \boldsymbol{x} + \int_0^{t/\alpha} \alpha\boldsymbol{v}^{\boldsymbol{\theta}}(\boldsymbol{\phi}^{\alpha\boldsymbol{\theta}}(\boldsymbol{x},\tau))\,d\tau \tag{10}$$

$$= \boldsymbol{x} + \int_0^{t} \boldsymbol{v}^{\boldsymbol{\theta}}(\boldsymbol{\phi}^{\alpha\boldsymbol{\theta}}(\boldsymbol{x},\eta/\alpha))\,d\eta = \boldsymbol{x} + \int_0^{t} \boldsymbol{v}^{\boldsymbol{\theta}}(\xi^{\boldsymbol{\theta}}(\boldsymbol{x},\eta))\,d\eta \tag{11}$$

and now observe that $\xi^{\boldsymbol{\theta}}(\cdot,t)$ coincides with $\boldsymbol{\phi}^{\boldsymbol{\theta}}(\cdot,t)$.

Part (ii). By applying the Picard-Lindelof theorem, and then extending the solution (to a finite t) we conclude that the trajectories do not intersect. From this it follows that $T^{\boldsymbol{\theta}}$ is invertible. Since $\boldsymbol{\theta} \mapsto \boldsymbol{v}^{\boldsymbol{\theta}}$ is linear, and since both $\mathbb{R}^d$ and $\mathcal{V}$ are linear spaces, we know that $\boldsymbol{v}^{-\boldsymbol{\theta}} \in \mathcal{V}$. Thus, by the definition of CPAB transformations, $T^{-\boldsymbol{\theta}} \in M$. We still, however, need to show that $T^{-\boldsymbol{\theta}} = (T^{\boldsymbol{\theta}})^{-1}$; *i.e.*, that $T^{-\boldsymbol{\theta}}$ is indeed the inverse of $T^{\boldsymbol{\theta}}$. By Eqn. (1) from the paper,

$$\boldsymbol{\phi}^{-\boldsymbol{\theta}}(\boldsymbol{\phi}^{\boldsymbol{\theta}}(\boldsymbol{x},t),t) = \boldsymbol{\phi}^{\boldsymbol{\theta}}(\boldsymbol{x},t) + \int_0^{t} \boldsymbol{v}^{-\boldsymbol{\theta}}(\boldsymbol{\phi}^{\boldsymbol{\theta}}(\boldsymbol{x},\tau))\,d\tau = \boldsymbol{\phi}^{\boldsymbol{\theta}}(\boldsymbol{x},t) - \int_0^{t} \boldsymbol{v}^{\boldsymbol{\theta}}(\boldsymbol{\phi}^{\boldsymbol{\theta}}(\boldsymbol{x},\tau))\,d\tau = \boldsymbol{x} \tag{12}$$

where we used the fact the map $\boldsymbol{\theta} \mapsto \boldsymbol{v}^{\boldsymbol{\theta}}$ is linear (so $\boldsymbol{v}^{-\boldsymbol{\theta}} = -\boldsymbol{v}^{\boldsymbol{\theta}}$).

Part (iii). Since we showed that $(T^{\boldsymbol{\theta}})^{-1} = T^{-\boldsymbol{\theta}}$ and that $T^{-\boldsymbol{\theta}} \in M$, it is enough to show that any $T^{\boldsymbol{\theta}} \in M$ is differentiable. This is a known result for transformations obtained by integration of (stationary) Lipschitz-continuous velocity fields, but the CPA structure enables us to outline another proof, specialized to this case. Since $\boldsymbol{v}^{\boldsymbol{\theta}}$ is continuous, it follows that $\{t_i(\boldsymbol{x},\boldsymbol{\theta},t)\}_{i=1}^{m_{\boldsymbol{x},\boldsymbol{\theta},t}}$ are continuous functions of $\boldsymbol{x}$. The fact that $m_{\boldsymbol{x},\boldsymbol{\theta},t}$ changes with $\boldsymbol{x}$ does not change it. It just means a nominal t_i decreases continuously to zero or increases continuously from zero. Thus, taking the derivative of $\boldsymbol{\phi}^{\boldsymbol{\theta}}(\boldsymbol{x},t)$ w.r.t. $\boldsymbol{x}$ results in summing terms of the form (notationally suppressing the dependency in t)

$$T_{\text{left},1}\,\frac{d\,\mathbf{expm}(t_i\widetilde{A_{c_i(\boldsymbol{x},\boldsymbol{\theta}),\boldsymbol{\theta}}})}{d\boldsymbol{x}}\,T_{\text{right},1} + T_{\text{left},2}\,\mathbf{expm}(t_i\widetilde{A_{c_i(\boldsymbol{x},\boldsymbol{\theta}),\boldsymbol{\theta}}})\,T_{\text{right},2} \tag{13}$$

where $T_{\text{left},1}$, $T_{\text{left},2}$, $T_{\text{right},1}$ and $T_{\text{right},2}$ are $(n+1)\times(n+1)$ matrices that change continuously with $\boldsymbol{x}$ (and also depend on i and $\boldsymbol{\theta}$). Likewise, the matrix $t_i\widetilde{A_{c_i(\boldsymbol{x},\boldsymbol{\theta}),\boldsymbol{\theta}}}$ changes continuously with $\boldsymbol{x}$. Thus, $\frac{d\,\mathbf{expm}(t_i\widetilde{A_{c_i(\boldsymbol{x},\boldsymbol{\theta}),\boldsymbol{\theta}}})}{d\boldsymbol{x}}$ is continuous too. $\square$

Part (iv). A well-known result is that in the case of an affine velocity field, $\boldsymbol{x} \mapsto A\widetilde{\boldsymbol{x}}$, a zero-trace A implies that, for any (finite) t, the resulting transformation,

$$\boldsymbol{x} \mapsto \begin{bmatrix} I_{n\times n} & \mathbf{0}_{n\times 1} \end{bmatrix} \mathbf{expm}(t\widetilde{A})\widetilde{\boldsymbol{x}}\,, \tag{14}$$

is volume preserving. From a Lagrangian standpoint, one way to show this is by noting that the determinant of the Jacobian matrix is one (since it is equal to the determinant of $\mathbf{expm}(t\widetilde{A})$ which is one since the trace is zero – a known property of $\mathbf{expm}$). From an Eulerian standpoint, this can be shown by applying the Divergence Theorem since a zero trace of A implies that the velocity field has zero divergence, and thus, by the Divergence theorem, no mass leaves or enters the region of interest (subject to a regularity condition on the boundary of the region). In the CPA case, if all the A's have zero trace, then it can be similarly shown that the determinant of the Jacobian is still one (also verified numerically). However, it is easier to prove this using the Eulerian approach. Any region (whose boundary satisfies the regularity condition of the Divergence theorem) can be divided to smaller subregions such that each subregion is fully contained within some cell. In each such cell, the divergence of the field is zero and the Divergence theorem is applicable and thus the net inward flux (of mass, but by taking the density to be 1, it means volume) equals the net outward flux. Since the total flux is zero in every cell, the same holds for the entire region and we conclude the transformation is volume preserving. While this proof is restricted to regions whose boundary is regular, one can appeal to continuity arguments (approximating any non-regular boundary from inside and outside using regular boundaries) to prove the more general case.

Part (v). Any space whose elements are invertible $\Omega \to \Omega$ maps is nonlinear, even if the maps themselves are linear[2]. To see that, note that the (constant) map, $\boldsymbol{x} \mapsto \mathbf{0}$, is the zero element of the linear space of all $\Omega \to \Omega$ maps (which contains M). This map is not invertible and is thus not in M. It follows that M is not linear since a linear subspace must contain the zero element. As for the dimension, intuitively this follows from the fact that M is defined via the d-dimensional $\mathcal{V}$. We omit the formal proof (which is based on using $\exp$ to create a chart) as this is a standard result in differential geometry.

[2] *E.g.*, rotations in 3D are linear maps but $SO(3)$ is nonlinear.

Part (vi). Let $\boldsymbol{v} : \Omega \to \mathbb{R}^n$ be differentiable and let $T : \Omega \to \Omega$ be given by $T(\boldsymbol{x}) = \boldsymbol{\phi}(\boldsymbol{x}, 1)$ where $\boldsymbol{\phi}(\boldsymbol{x}, t) = \int_0^1 \boldsymbol{v}(\boldsymbol{\phi}(\boldsymbol{x}, \tau))\, d\tau$. At every $\boldsymbol{x}$, $\boldsymbol{v}(\boldsymbol{x})$ can be approximated by a first-order Taylor expansion about $\boldsymbol{x}_0$ where $\boldsymbol{x}_0$ is the, say, first vertex of $U_{\gamma(\boldsymbol{x})}$. Let $\varepsilon > 0$. By picking $\mathcal{P}$ fine enough, we can always find $\boldsymbol{v}_{\boldsymbol{A}} \in \mathcal{V}'$ such that $\max_{\boldsymbol{x}\in\Omega} ||\boldsymbol{v}(\boldsymbol{x}) - \boldsymbol{v}_{\boldsymbol{A}}(\boldsymbol{x})|| < \varepsilon/2$ (where we absorbed the constant $\boldsymbol{v}(\boldsymbol{x}_0)$ into $\boldsymbol{v}_{\boldsymbol{A}}$) and that $\max_{\boldsymbol{x}\in\Omega} ||\boldsymbol{v}_{\boldsymbol{A}}(\boldsymbol{x}) - \boldsymbol{v}^{\boldsymbol{\theta}}(\boldsymbol{x})|| < \varepsilon/2$, where $\boldsymbol{\theta} = B^T \mathbf{vec}(\boldsymbol{A})$. It follows that $\max_{\boldsymbol{x}\in\Omega} ||\boldsymbol{v}(\boldsymbol{x}) - \boldsymbol{v}^{\boldsymbol{\theta}}(\boldsymbol{x})|| < \varepsilon$. Consequently, we can approximate $\boldsymbol{v}$ arbitrarily well. Since $\boldsymbol{v} \mapsto T$ is continuous, this implies that we can also approximate T arbitrarily well.

Proof of Theorem 2. In the notation below we drop the dependency in $\boldsymbol{\theta}$. Recall $n = 1$. Thus, we can write $A_c = [a_c, b_c]$, and $\widetilde{A_c} = \left[\begin{smallmatrix} a_c & b_c \\ 0 & 0 \end{smallmatrix}\right]$. Also let $U_c = [x_c^{\min}, x_c^{\max}]$ denote the c^{th} interval. For 2×2 matrices the matrix exponential is given in closed form [3]. If in addition, as we have here, the last row contains only zeros, this solution is given by

$$\mathbf{expm}\left(\begin{bmatrix} ta_c & tb_c \\ 0 & 0 \end{bmatrix}\right) = \begin{cases} \begin{bmatrix} e^{ta_c} & \frac{b_c(e^{ta_c}-1)}{a_c} \\ 0 & 1 \end{bmatrix} & \text{if } a_c \neq 0 \\[2ex] \begin{bmatrix} 1 & tb_c \\ 0 & 1 \end{bmatrix} & \text{if } a_c = 0 \end{cases} \tag{15}$$

where we assumed $t \neq 0$ (if $t = 0$ then we have $\exp(\mathbf{0}_{2\times 2}) = I_{2\times 2}$). Thus,

$$\psi_c^t(\boldsymbol{x}) \triangleq \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{expm}\left(\begin{bmatrix} ta_c & tb_c \\ 0 & 0 \end{bmatrix}\right) \begin{bmatrix} \boldsymbol{x} \\ 1 \end{bmatrix} = \begin{cases} e^{ta_c} x + \frac{b_c(e^{ta_c}-1)}{a_c} & \text{if } a_c \neq 0 \\[2ex] x + tb_c & \text{if } a_c = 0 \end{cases} \tag{16}$$

(note this includes the case of $t = 0$). Without loss of generality, suppose $t > 0$. Let $\gamma(\boldsymbol{x}) = c_1$ be the index of the interval containing $\boldsymbol{x}$. Now suppose $\boldsymbol{v}(\boldsymbol{x}) > 0$; i.e., the direction of the velocity at location $\boldsymbol{x}$ is rightward. Let $t_1(\boldsymbol{x})$ denote the time it takes to hit the right boundary of $U_{c_1} = [x_{c_1}^{\min}, x_{c_1}^{\max}]$:

$$t_1 \triangleq \min\{t : \psi_{c_1}^t(\boldsymbol{x}) = x_{c_1}^{\max}\} \tag{17}$$

where, by convention, the minimum of an empty set is ∞. Solving for t_1,

$$t_1 = \infty \text{ if } \boldsymbol{v}(x_{c_1}^{\max}) \leq 0; \text{ otherwise: } t_1 = \begin{cases} \frac{1}{a_{c_1}} \log\left(\frac{x_{c_1} + \frac{b_{c_1}}{a_{c_1}}}{x + \frac{b_{c_1}}{a_{c_1}}}\right) & \text{if } a_{c_1} \neq 0 \\[2ex] \frac{x_{c_1} - x}{b_{c_1}} & \text{if } a_{c_1} = 0 \end{cases} . \tag{18}$$

If $t_1 > t$, then $\boldsymbol{\phi}^{\boldsymbol{\theta}}(\boldsymbol{x}, t)$ is simply $\psi_{c_1}^t(\boldsymbol{x})$. Otherwise, it means we moved to next interval to the right, U_{c_1+1}, and we need to redo the process with $x_{c_1}^{\max}$ as the new starting point and $t - t_1$ instead of t. Since t is finite, this process will necessarily converge in a finite amount of steps, $m_{\boldsymbol{x},t}$. A loose upper bound on $m_{\boldsymbol{x},t}$ is $N_c - c_1 + 1$. The case where $\boldsymbol{v}(\boldsymbol{x}) < 0$ is handled similarly. Taken together,

$$\boldsymbol{\phi}(\boldsymbol{x}; t) = (\psi_{c_m}^{t_m} \circ \ldots \circ \psi_{c_2}^{t_2} \circ \psi_{c_1}^{t_1})(\boldsymbol{x}) \tag{19}$$

where for $2 \leq i \leq m$, $c_i = c_{i-1} + \text{sign}(\boldsymbol{v}(\boldsymbol{x}))$, while m and $\{t_i\}_{i=1}^m$ are found by the aforementioned procedure. □

4. Accuracy of the Proposed Solver and Details about the Generic Solver used as its Subroutine

Recall that our solver has two parameters, N_{steps} and n_{steps}. These imply two step sizes, one large, $\Delta_t = t/N_{\text{steps}}$, and one small $\delta_t = \Delta_t/n_{\text{steps}}$. The reason we can make large (and thus fewer) steps is that we use an exact analytic update when possible. Now, the astute reader might by concerned at this point that by doing several consecutive analytic updates, we effectively take powers of $T_{A_{c,\boldsymbol{\theta}}} = \mathbf{expm}(\Delta_t \widetilde{A_c})$. While mathematically, $(\mathbf{expm}(\Delta_t \widetilde{A_c}))^k = \mathbf{expm}(k\Delta_t \widetilde{A_c})$, on a computer numerical errors accumulate and the equality may not hold; see [4, 1] for a more in-depth treatment of such numerical issues. In general, this potential problem is most noticeable when the matrices are full (as opposed to sparse), when they are large, and when many powers are taken. However, in our case, this issue does not arise in practice since our matrices are small (since $n \in 1, 2, 3$), their last row is all zeros, and we take only a small number of steps.

Recall that our algorithm uses a generic solver as its subroutine. Additionally, we also use the same generic solver for timing (in the paper) and accuracy (here) comparisons. Particularly, the generic solver we chose is the modified Euler Method. This solver requires the specification a time step, as well as number of steps. For comparison purposes, the generic solver was run with step size δ_t and $N_{\text{steps}} \times n_{\text{steps}}$ steps. Since our solver uses less steps, it is faster than the generic solver, as was also verified empirically in the paper. Since instead of many approximated steps it uses fewer exact steps, it is also more accurate. We verify this empirically, by drawing for each value of N_c, 100 random CPA fields form the prior, $\{\boldsymbol{v}^{\boldsymbol{\theta}_i}\}_{i=1}^{100}$, and sampling 1000 points in Ω, $\{x_j\}_{j=1}^{1000}$, and then computing the error,

$$\epsilon = \frac{1}{100}\sum_{i=1}^{100}\frac{1}{1000}\sum_{i=j}^{100}\left|T^{\boldsymbol{\theta}}_{\text{exact}}(x_j) - T^{\boldsymbol{\theta}}_{\text{solver}}(x_j)\right|. \tag{20}$$

Table 1 contains the relative improvements, $\frac{\epsilon_{\text{generic}} - \epsilon_{\text{ours}}}{\epsilon_{\text{generic}}}$, for different values of N_c.

N_c	5	25	45	65	85	105
improvement [%]	2.99	12.56	11.26	9.98	3.29	3.73

Table 1: Accuracy: Average improvement, in %, of our proposed solver with $N_{\text{steps}} = 10$ and $n_{\text{steps}} = 10$ over the generic solver with 100 steps.

5. Some Differential-Geometric Aspects

Readers uninterested in differential geometry may safely skip this section. As many readers must have already recognized, $\mathcal{V}$ is the tangent space at the identity to M. Just like the transformation space from [2], M is not closed under composition and thus it not a Lie group; equivalently, $\mathcal{V}$ is not a Lie algebra (even though it can be viewed as a linear subspace of several Lie algebras) due to lack of closure under the Lie bracket. Note that $\mathcal{V}$ is a finite-dimensional linear subspace of $T_I G$, the infinite-dimensional Lie algebra of the Lie group of $\Omega \to \Omega$ diffeomorphisms The map $\exp : \mathcal{V} \to M$ is the restriction of the Lie group exponential map, $\exp : T_I G \to G$, to $\mathcal{V} \subset T_I G$. This approach, of restricting the exponential map of a Lie group to a linear subspace of the Lie algebra when the subspace lacks a Lie-bracket closure, has been found useful, *e.g.*, in applications involving Symmetric Positive-Definite (SPD) matrices[3]; *e.g.*, [5].

Unlike the inner product on $\mathcal{V}$ mentioned in the paper, the inner product (again, on $\mathcal{V}$) given by

$$\langle\cdot,\cdot\rangle_{\text{vert}} : (\boldsymbol{v}^{\boldsymbol{\theta}_1}, \boldsymbol{v}^{\boldsymbol{\theta}_2}) \mapsto \textstyle\sum_{\boldsymbol{\xi}:\boldsymbol{\xi}\in\mathcal{P}_{\text{vert}}} \boldsymbol{v}^{\boldsymbol{\theta}_1}(\boldsymbol{\xi}) \cdot \boldsymbol{v}^{\boldsymbol{\theta}_2}(\boldsymbol{\xi}). \tag{21}$$

is close in spirit to the oft-used inner product on $T_I G$:

$$\langle \boldsymbol{v}_1, \boldsymbol{v}_2 \rangle = \int_\Omega \boldsymbol{v}_1(\boldsymbol{\xi}) \cdot \boldsymbol{v}_2(\boldsymbol{\xi})\, d\boldsymbol{\xi} \quad \boldsymbol{v}_1, \boldsymbol{v}_2 \in T_I G. \tag{22}$$

As we pointed out in the discussion on smoothness above, one of the advantages of the inner product we used in the paper over the inner product above is related to eliminating fictitious noise.

Another difference between defining priors using B versus defining priors using the vertex-based basis, and this difference can be either a plus or a minus depending on the application, is that B depends on choice of the origin of Ω while the vertex-based basis does not.

In this work we adopted an approach that, at least implicitly, exploits (the restriction of) the Lie group exponential associated with G. Future work should explore Riemannian geometries of M as well as their connection to those of G. While one can define Riemannian metrics that are specific to M, it is worth noting that since M is a submanifold of G, the former may also inherit any Riemannian metric from the latter.

[3] SPD is not a matrix Lie group since it is not closed under composition (at least not w.r.t. the standard group binary operation of matrix Lie groups; *i.e.*, matrix multiplication). Equivalently, its tangent space at the identity, the space of symmetric matrices, is not closed under the Lie Bracket.

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